

Task NO: 7
Date: 23/9/25

PL/SQL Procedures and Functions

Aim: To implement PL/SQL procedures, functions and loops on number theory and business scenarios.

Procedure:

PL/SQL is a combination of SQL along with the procedural features of programming languages. It was developed by Oracle Corporation in the early 90's to enhance the capabilities of SQL. PL/SQL is one of three key programming languages embedded in Oracle database along with SQL itself and Java.

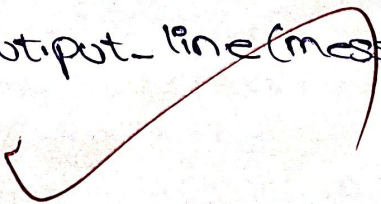
Simple problem to print a sentence:

Syntax:

```
DECLARE
    <declaration section>
BEGIN
    <executable section>
Exception
    <exception handling>
END;
```

Program:

```
DECLARE
    message varchar2(20) := 'booking closed';
BEGIN
    dbms_output.put_line(message);
END;
```



Dynamic Input:

Set Serveroutput on;
declare

X number(5);

Y number(5);

Z number(9);

begin

X := 10;

Y := 12;

Z := X+Y;

dbms_output.put_line('Sum is '||Z);

end;

/

Output: Sum is 22

declare

var1 integer;

var2 integer;

var3 integer;

begin

var1 := &var1;

var2 := &var2;

var3 := var1+var2;

dbms_output.put_line(var3);

end;

/

Enter values for var1: 20

old 6: var1 := &var1;

new 6: var1 := 20;

Enter value for var2: 30

old 7: var2 := &var2;

new 7: var2 := 30;

50

Declare

hid Number(8); = 100;

Begin

if (hid = 10) then

dbms_output.put_line('value of hid is 10');

else if (hid = 20) then

dbms_output.put_line('value of hid is 20');

else if (hid = 30) then

dbms_output.put_line('value of hid is 30');

else

dbms_output.put_line('None of the value is matching');

End if;

dbms_output.put_line('Exact value of hid is: '||hid);

End;

/

Output:-

None of the values is matching

Exact value of hid is: 100

Declare

hid number(i);

oid number(i);

Begin

<< outer_loop >>

for hid in 1..3 loop

<< inner_loop >>

for oid in 1..3 loop

dbms_output.put_line('hid is: '||hid||' and is: '||oid);

END loop inner_loop;

END loop outer_loop;

END;

/

hid is : 1 and oid is : 1

hid is : 1 and oid is : 2

hid is : 1 and oid is : 3

hid is : 2 and oid is : 1

hid is : 2 and oid is : 2

hid is : 2 and oid is : 3

hid is : 3 and oid is : 1

hid is : 3 and oid is : 2

hid is : 3 and oid is : 3

Program for only procedure :

create or replace procedure cs information

(c_id in number, c_name in varchar2)

is

begin

dbms_output.put_line('ID: ' || c_id);

end;

|

Procedure created

exec cs information(101, 'ram');

PL/SQL procedure successfully completed

set serveroutput on;

exec cs information(101, 'ram');

ID: 101

Name: ram.

Program for only functions

create or replace function cs information

(h_id in number, c_name in varchar2)

Return varchar2

is

begin

if (c_id > 200 then)

Return ('no booking available');

False

Return ('booking open');

end if;

end;

Function created.

declare

msg varchar2 (200);

begin

msg := cs information 2 (102, 'dam');

dbms_output.put_line (msg);

end;

|

Vehicle - available

declare

merge varchar2 (200);

begin

merge = cs information 2 (200, 'dam');

dbms_output.put_line (merge);

end;

|

No vehicle available.

VELTECH	
EX No.	7
PERFORMANCE (5)	5
RESULT AND ANALYSIS (3)	5
VIVA VOCE (3)	5
RECORD (4)	5
TOTAL (15)	15
SIGN WITH DA	


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Result: Thus the implementation of PL/SQL procedures, functions and loops on number theory and business scenario was completed.

Aim: To implement PL/SQL programs using loops for printing prime number customers IDs and for demonstrating loop control in different scenarios.

Procedure:

1. Start a PL/SQL block or procedure.
2. Use a cursor (if required) to fetch customer IDs from a table.
3. For each ID, check whether it is a prime number using a loop.
4. Use for loop/while loop to demonstrate prime number checking.
5. Print the result using `dbms_output.put_line`.
6. End the block.

Using while loop with cursor

Create or replace procedure print_prime_customers is.

Cursor cust_cur is

Select customer_id from customers;

v_id number;

v_is_prime Boolean;

v_i number;

Begin

open cust_cur;

loop

fetch cust_cur into v_id;

exit when cust_cur % not found;

If v_id < 2 then

v_is_prime := false;

else v_is_prime := true;

v_i := 2;

while v_i <= sqrt(v_id) loop.

IF V-is-prime THEN

DBMS-output.put-line ('Prime : ' || V-num);

V-Count := V-Count + 1;

END IF;

V-num := V-num + 1;

END loop;

END;

VELTECH	
EX No.	27
PERFORMANCE (5)	9
RESULT AND ANALYSIS (3)	9
VIVA VOCE (3)	5
RECORD (4)	5
TOTAL (15)	15
SIGN WITH DATE	


b/w

Result: Thus the implementation of PL/SQL procedures and functions and loops on number theory and business scenario was completed.